DIFFERENTIAL TOPOLOGY

Contents

1. Poincaré-Hopf theorem

Definition 1.1. Let $x \in X$ such that $V(x) = 0$ for some $V \in \mathfrak{X}(X)$. x is an *isolated zero* of V is there is an open set $U \ni x$ such that for every $y \in U \setminus \{x\}$, $V(y) \neq 0$.

Definition 1.2.

$$\operatorname{ind}(V, x) = \operatorname{wind}(\tilde{V}|_{\partial B_r(0),0} \overset{\text{def}}{=\!=} \left(\frac{\tilde{V}}{\|\tilde{V}\|}\Big|_{\partial B_r(0)} \right)$$

Theorem 1.3 (Poincaré-Hopf)**.** *Let X be a compact manifold without boundary and $V \in \mathfrak{X}(X)$ with isolated zeroes. Then*

$$\sum_{x \in X} ind_x = \chi(X)$$

2. Hopf's theorem

Theorem 2.1 (Hopf)**.** *Let X be a compact, oriented and connected manifold (without boundary). Let $f, g : X \to S^n$. (Notice degree is well-defined because X is compact and S^n is connected.) Then*

$$f \sim g \iff degf = degg$$

Which says that

References